

Everywhere = Nowhere = Now

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GitHub: https://github.com/docomomomo/CART360_MainProject

Abstract

TalkGlove

This project explores how wearable technology can be used to support communication in environments that are heavily built around spoken language, focusing on a gesture-touch based communication glove that uses capacitive touch sensors to detect finger inputs which are then translated into a modular vocabulary system. These inputs are processed through a microcontroller and output as both text and speech allowing communication to happen directly through the body instead of relying on external devices like phones or keyboards.

The project comes from the idea that most daily interactions assume that everyone can speak, which often creates barriers for people who are deaf, nonverbal or have speech impairments. This led to exploring how interaction could be embedded into something wearable so that communication feels more immediate, personal and integrated into everyday movement. Over time, the project developed through multiple sensing approaches and technical iterations, gradually shifting toward a system that feels more fluid, responsive and natural to use.

Instead of trying to replicate complex systems like sign language translation, the project focuses on simplified and customisable gestures that prioritise ease of use and adaptability, allowing users to communicate in a way that works for them. In this sense the glove is not just a tool, but something that sits between people and helps translate intention into communication opening up different ways for interaction and expressing ideas.

Design Narrative

This project began with the intention of exploring how communication could move away from systems that rely heavily on spoken language with a focus on creating a wearable interface that could translate physical gestures into speech, allowing users to communicate through the body rather than relying on external devices like phones or text based systems. The idea came from recognising how many everyday environments are built around the assumption that everyone can speak and be understood in the same way, which in practice creates small but constant barriers for people who are

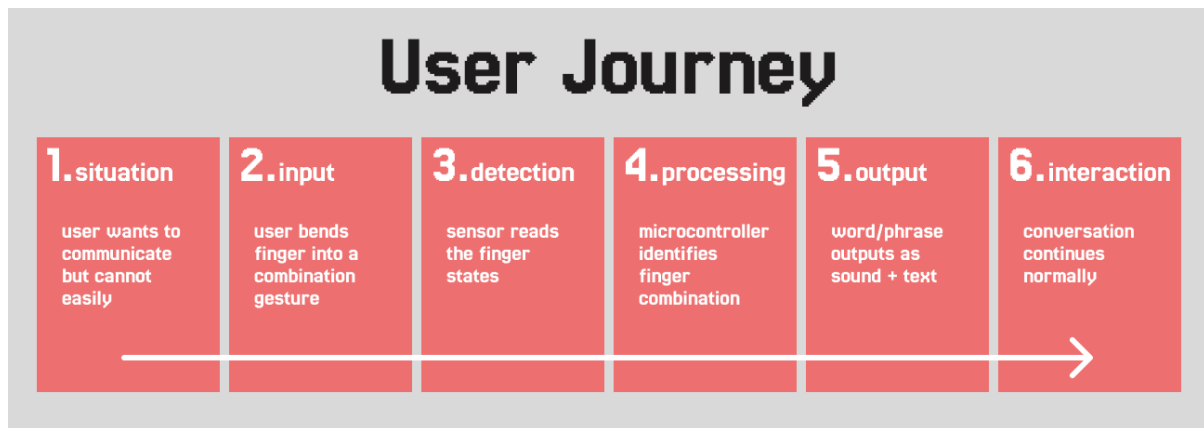
nonverbal or have speech impairments. This led to questioning that assumption and exploring whether communication could instead be embedded into the body through a tangible interface, where interaction happens more naturally as part of movement rather than through a separate tool. Existing systems like the Ava app and projects such as SignAloud Gloves show how technology can translate communication across different forms, but they also highlight how many of these systems still rely on predefined structures or external devices.

As the project developed the focus shifted from simply translating gesture into speech to thinking more carefully about how that interaction actually feels to use, especially in terms of comfort, responsiveness and how naturally it fits into everyday behaviour. Instead of trying to replicate complex systems like sign language or reinvent the wheel completely, the design moved toward simplified and modular gestures that prioritise speed, accessibility and adaptability, allowing users to build communication in a way that suits their own movement and ability. This shift reframed the glove not as something that interprets language in a fixed way, but as something that supports expression in a more flexible way.

Through this approach, interaction is no longer tied to a specific device or moment but instead becomes something that is embedded within the body and available at any time, where the glove acts as a system that is always present and ready to be used without needing to pick up or navigate anything else. The project explores how technology can move from something we use occasionally to something we exist alongside, where communication becomes more immediate and continuous rather than delayed or interrupted.

The design ultimately positions the glove as something that sits between physical gesture and digital communication, acting as a facilitator rather than a replacement for existing systems. It is not trying to replace sign language or other assistive technologies, but instead offers an alternative that is more immediate, customisable, and physically integrated. Through this, the project reflects a broader interest in how wearable technology and physical computing can reshape the way people interact, communicate and connect with others, especially when the system is designed to adapt to the user rather than forcing the user to adapt to it. Projects like the SoundShirt also show how wearable systems can embed interaction directly into the body, turning experience into something that is felt and lived through rather than accessed externally, which further supports the idea of technology becoming part of how we naturally communicate and engage with the world

User Journey Map



Prototype Process

The prototyping phase of the project focused on exploring different ways of detecting finger input, where each iteration revealed both technical and physical limitations that gradually shaped the direction of the final design.

The initial approach used flex sensors, similar to those found in systems like the Nintendo PowerGlove, which allowed the system to detect how much each finger was bending and aligned closely with the original idea of capturing natural hand movement. While this supported a more organic form of interaction, the flex sensors quickly showed clear issues as they were extremely delicate and not reliable for repeated use and at the same time limited how freely the hand could move. This created a contradiction in the design, where something that was meant to feel natural and seamless instead started to restrict movement and introduce new difficulties for the user. Because of this, the design shifted toward using mechanical push buttons embedded into the glove, as a way to prioritise reliability and simplify the input system through clear binary states. While this worked well from a technical point of view, it introduced a different set of problems in terms of physical experience. The buttons added noticeable bulk, especially in the palm, which made the glove uncomfortable to wear and awkward to use and the interaction began to feel forced rather than natural, as users had to press against rigid components instead of moving freely. Instead of supporting communication through the body, the glove started to feel intrusive and physically disruptive.

Multiplexers were explored as a way to expand the number of inputs, particularly when working with flex sensors, but this added another layer of complexity to the system that did not meaningfully improve the interaction. The additional wiring and setup made the system harder to manage and introduced more points where things could go wrong, which slowed down development without actually improving the overall experience.

These issues led to a reconsideration of the sensing approach, shifting focus toward finding a solution that could better support the intended interaction without adding unnecessary complexity. This is where the project moved to testing the MPR121 capacitive touch sensor. Unlike the previous methods, the MPR121 allowed the glove to detect touch-based input instead of relying on bending or pressure which made the interaction feel much more fluid and responsive, as finger contact could be used as a trigger without restricting movement or adding bulk to the design.

Through this iteration, the interaction became much more aligned with the original intention of the project, as the capacitive sensing approach simplified the system while also improving usability, comfort and responsiveness. Because of this the MPR121 was adopted as the main input system for the final artifact, shifting away from rigid and constrained interaction toward something that feels more natural, lightweight and integrated into the body.

Final Artifact

The final artifact takes the form of a modular, wearable communication system that translates gesture-based touch input into text and speech, built around a Raspberry Pi Pico 2 W microcontroller that processes input from an MPR121 capacitive touch sensor. These touch inputs are mapped to a custom vocabulary system, allowing users to trigger words through combinations of finger contact, which turns simple gestures into a way of constructing language.

The hardware setup includes a 1602 I2C LCD display for visual output, a speaker connected through an audio converter and amplifier for sound output, and a microSD card reader intended for storing and playing audio files, all connected through a breadboard circuit that supports rapid prototyping and ongoing iteration. This setup allows the system to be easily modified and adjusted, but also reflects its current state as a working prototype rather than a fully resolved product.

The system is designed in a modular way, where the interaction components are separated from the main processing hardware, with the glove containing the capacitive touch inputs, display, and speaker, while the Pico and supporting circuitry remain on the breadboard. This separation makes it easier to transport, modify, and test different parts of the system without needing to rebuild everything each time, allowing the project to develop more flexibly over time.

The interaction itself is designed to be simple and direct, where the user communicates by pressing combinations of touch inputs on the glove, which are then read by the microcontroller and translated into predefined words within the vocabulary system. By combining different inputs, users are able to build short phrases and sentences, which

are then output visually on the display and, when working, through audio playback, allowing communication to happen across both text and sound.

In its current state, the core interaction between input and output is working reliably, as the system is able to detect finger combinations and translate them into readable text in real time, clearly demonstrating the main idea of the project. The audio system has been implemented and coded, but is not fully functional due to compatibility issues with the microSD card, which means that sound playback is not present in the final artifact. Despite this, the text-based output is still effective in showing how the system is intended to work.

Overall, the final artifact functions as a working proof of concept that shows how gesture-based input can be translated into language through a wearable interface, and while it is still in a prototypical state, it establishes a clear foundation for further development, refinement, and improvement in both interaction and physical design.

Observations

The final artifact is able to demonstrate the core idea of the project, but reveals clear limitations in both the interaction and the physical design, showing a gap between a working system and a fully resolved artifact. The strongest part of the project is the interaction between finger input and text output, where the use of capacitive touch creates a more fluid and responsive experience compared to earlier prototypes, allowing the system to reliably detect combinations of inputs and translate them into words. This shows that the main idea of using gesture-touch based inputs as a way to trigger language is functional and that communication through simple input systems can work without relying on traditional interfaces like keyboards or screens.

Even though the input system works well, the overall experience still feels incomplete especially because the audio output which was intended to be a key part of the interaction did not function due to compatibility issues with the microSD card. This limits the system's ability to operate in real world situations, as communication is currently restricted to text only and as a result the artifact does not fully achieve its goal of providing a dual mode communication system through both sound and text.

A more significant issue appears in the materiality and physical construction of the glove, where the use of a latex glove and externally attached components results in something that feels temporary, fragile and unpolished. The wires, speaker and display are visibly attached in a way that lacks cohesion, making the artifact feel more like a rush job of parts rather than a completed system, which directly affects how the interaction is experienced. As discussed in the E-Acrylic research (Han et al., 2024), embedding electronics directly into the material changes how an interactive system is

experienced, shifting it from a collection of separate components into something more unified where the structure, interaction and appearance are all connected. In this project, the lack of that integration creates a disconnect between the idea of a seamless wearable interface and the actual experience of using it.

This also highlights how material choices are not separate from interaction but directly shape how a system is understood and used, where the physical construction influences both comfort and usability. The current glove being uncomfortable and difficult to wear, makes the interaction feel less natural, which goes against the original goal of embedding communication into the body. Instead of disappearing into the user's movement, the artifact becomes something that is constantly felt and noticed.

The decision to prioritise rapid prototyping made it easier to test and iterate on the system, but it also meant that less time was spent resolving the final form resulting in a clear gap between the concept and its physical execution. While the system works on a technical level, it does not yet feel like something that could exist naturally in everyday life. This outcome was also shaped by practical constraints around time, materials, and access to tools, which meant that the artifact had to be designed in a way that could be easily assembled and taken apart rather than permanently built. Because the glove and its components could not be kept in a permanent state, the system was intentionally made modular and removable, allowing it to be quickly disassembled. While this approach supported iteration and made the project manageable within these constraints, it also affected the overall finish and cohesion of the final artifact. In hindsight, these limitations could have been planned for more carefully as a lack of consideration around how to work within these constraints contributed to the final result feeling less resolved than intended.

Portability is another limitation as the main components being housed on a breadboard restrict the system to a controlled setup and make it difficult to use outside of that context. Without a compact form or protective casing, the artifact does not align with its intention of being a wearable device that can be used freely in different environments.

The project succeeds in demonstrating how gesture-touch based input can be translated into language, but it falls short in delivering a fully integrated and wearable experience. These shortcomings make it clear that achieving a functional system is only one part of the process and that equal attention needs to be given to how materials, form and electronics are brought together to create a cohesive and usable artifact.

Future Directions

Future development of the project would focus on improving both the physical construction of the glove and the overall usability of the system, with the aim of moving it closer to something that could be used naturally in everyday situations rather than remaining as a prototype.

A key area for improvement is the materiality of the wearable itself, where moving away from a latex based prototype toward a more durable, cloth based glove would make the artifact more comfortable, wearable and practical to use over longer periods of time. A fabric based approach would also allow the electronic components to be more properly integrated into the glove rather than being attached externally, creating a more cohesive system where the structure and interaction feel connected rather than separate.

Another important step would be developing proper casing and housing for the electronic components, as the current breadboard setup exposes the circuitry and limits portability making the system difficult to use outside of a controlled environment. Designing compact enclosures for the microcontroller and supporting hardware would not only make the system more robust and easier to carry but would also improve the overall visual quality of the artifact, helping it feel more like a resolved object rather than a temporary setup.

Further refinement would focus on improving the reliability and completeness of the interaction system, particularly by resolving the audio output issues so that the glove can fully function as a dual mode communication system through both sound and text. Expanding and refining the vocabulary system would also allow for more flexible and natural communication, giving users the ability to express a wider range of ideas through combinations of inputs.

Beyond technical improvements, the project could also be developed further as a more personalised communication system, where users are able to customise gestures more easily without needing to modify code. This would make the system more accessible to people with different needs and levels of mobility allowing the interaction to better adapt to the user rather than requiring the user to adapt to the system.

Overall, future work would focus on bringing together the interaction, materials and hardware into a more unified design so that the glove feels like a natural extension of the body rather than something separate. This includes improving comfort, portability and visual finish, while also addressing the limitations identified in the current prototype allowing the project to move from a working system toward a more complete and usable wearable artifact.

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